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2024-CET-04

Department:

Software Engineering

Group:

Computer Engineering Technology

Semester:

4th Semester

Subject:

Computer Network

Lab Task:

Create and Test a Simple LAN in Cisco Packet Tracer

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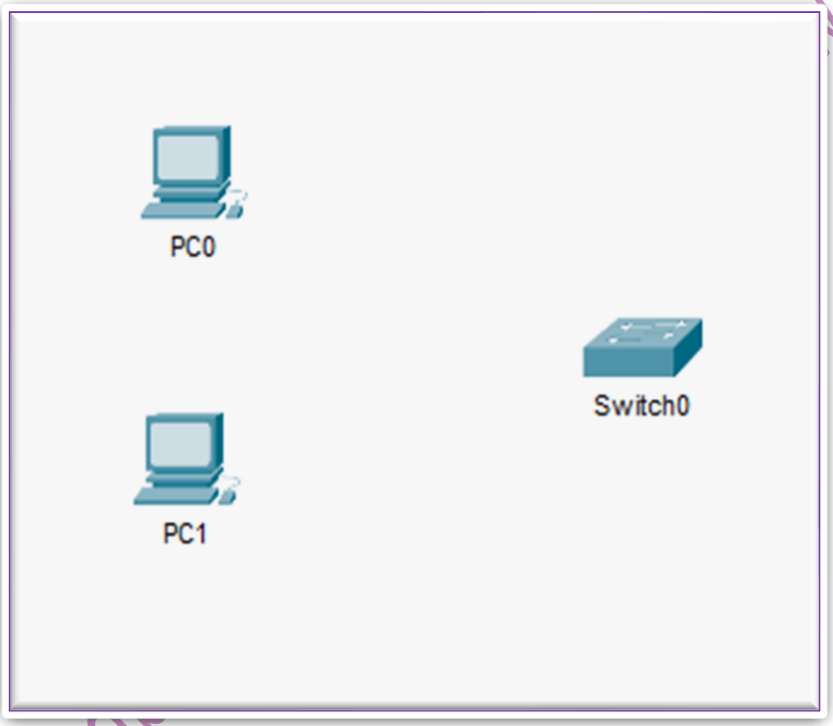
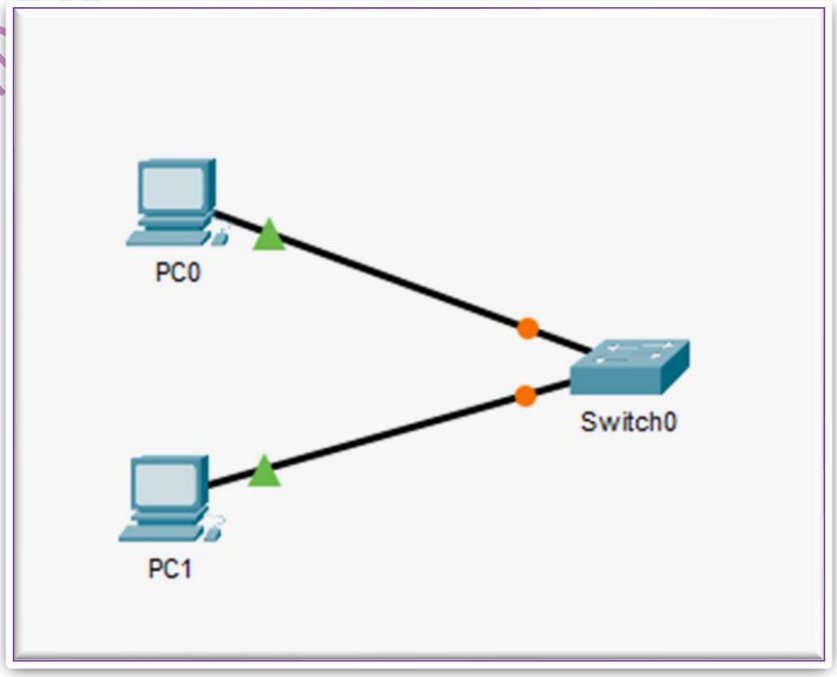
Lab Task 1

Network Topology Creation

1 Switch and 2 PC



Connect PC0 and PC1 with Switch



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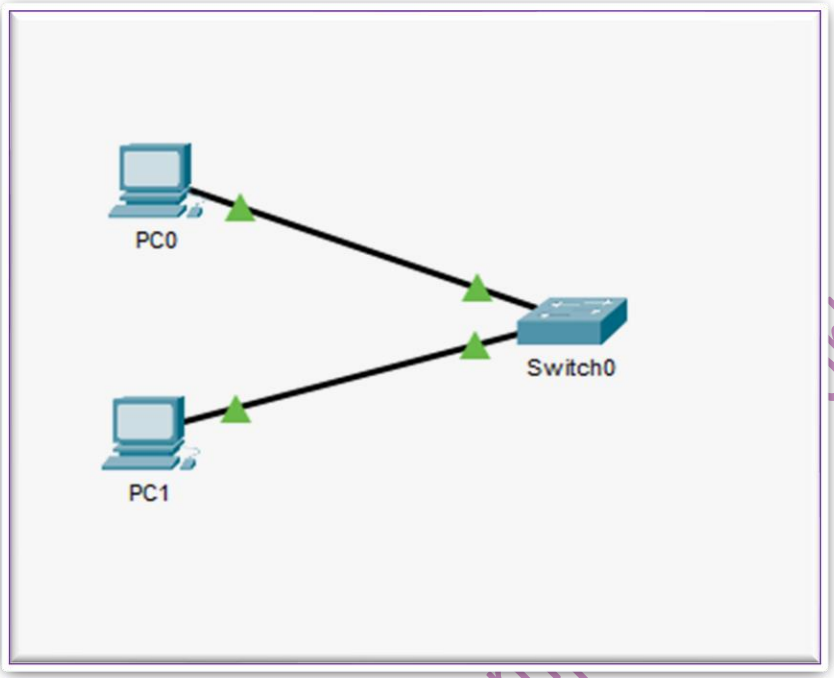
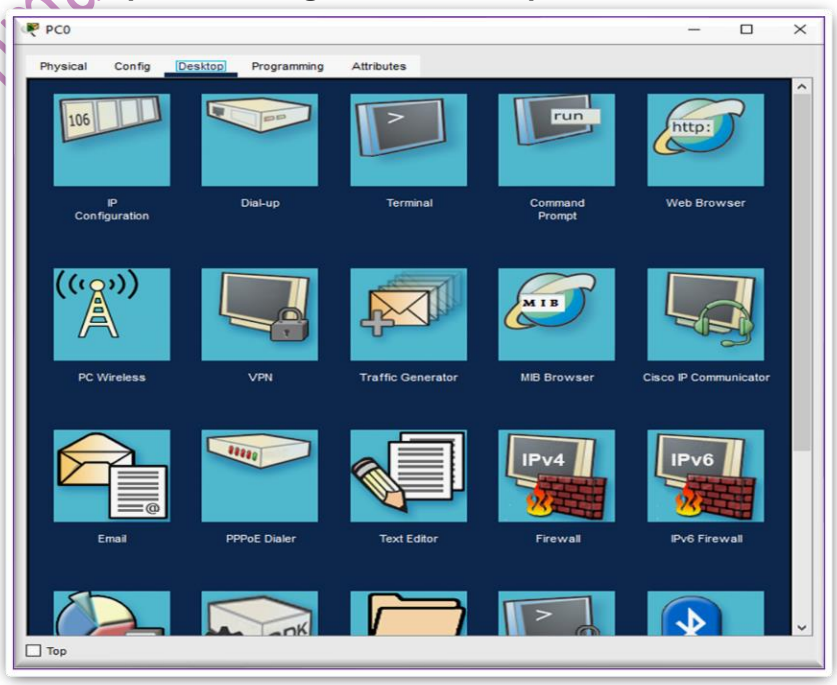


We connected two PCs with switch as shown in the figures. And PCs successfully connected to the switch with green light indicators.

IP Address Configuration

Open IP Configuration in first pc "PC0"



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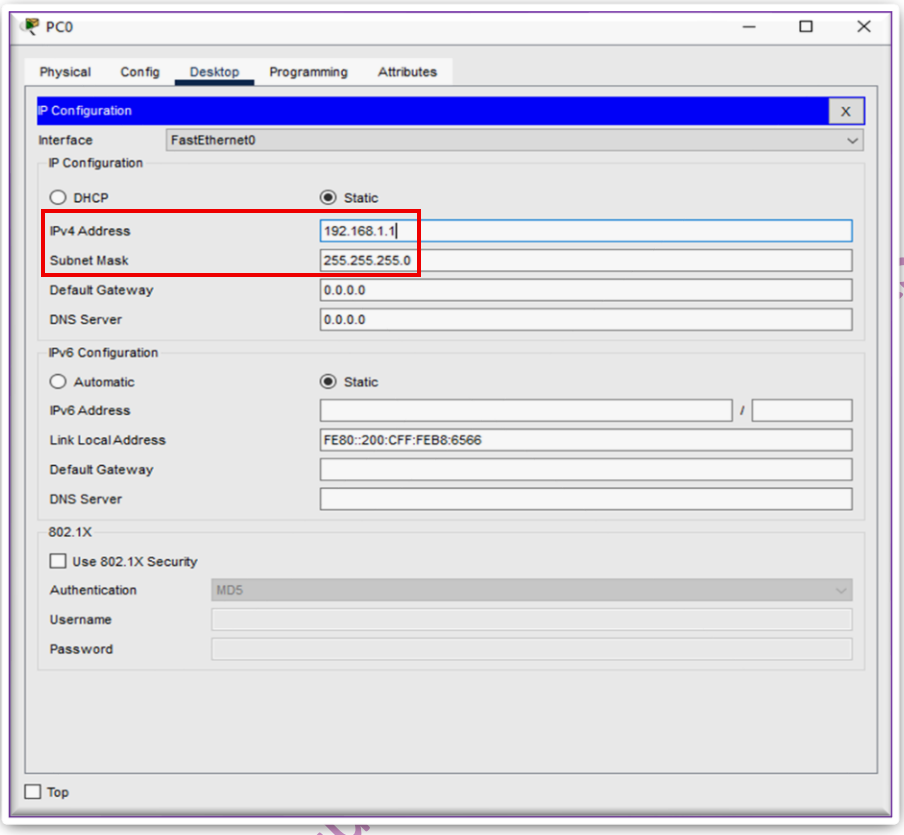
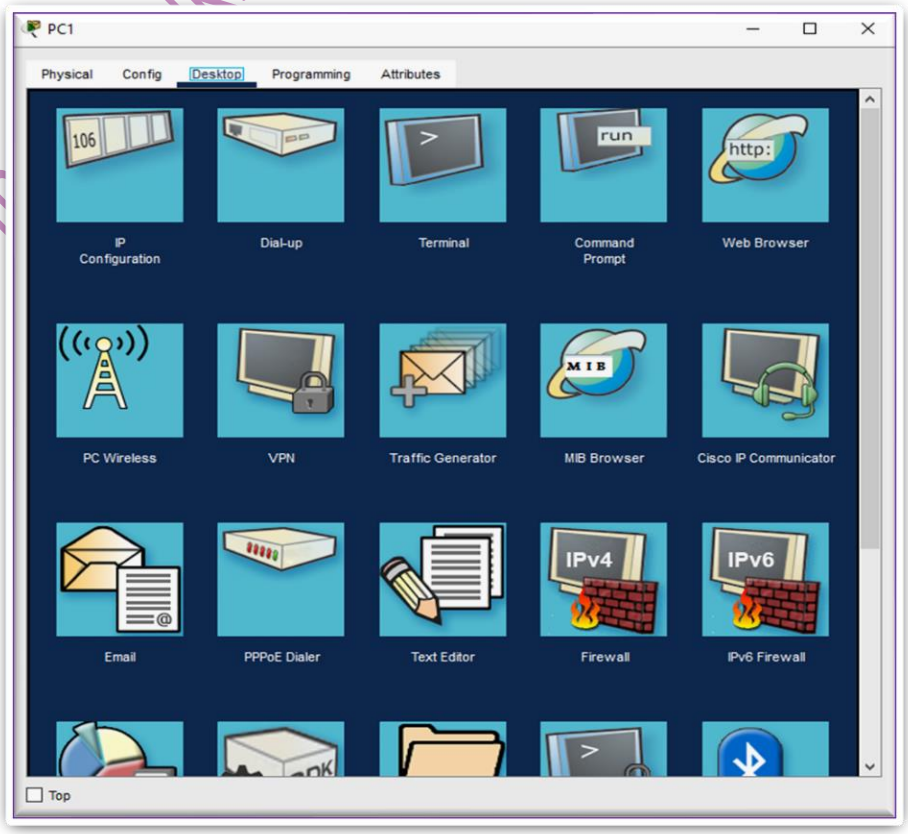
Set IP Address and Subnet Mask for PC 1



Open IP Configuration in Second pc "PC1"



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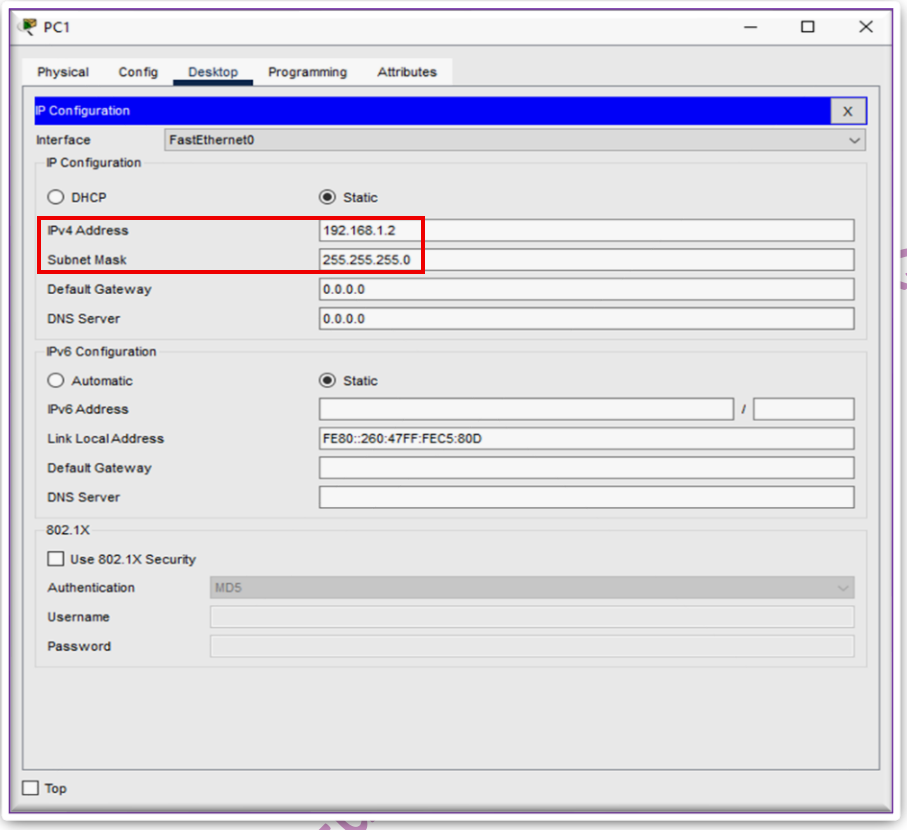
Set IP Address and Subnet Mask for PC 1



In this section we configure IP addresses manually for both PCs "PC0" and "PC1".

Both PCs have different IP addresses:

PC0: 192.168.1.1

PC1: 192.168.1.2

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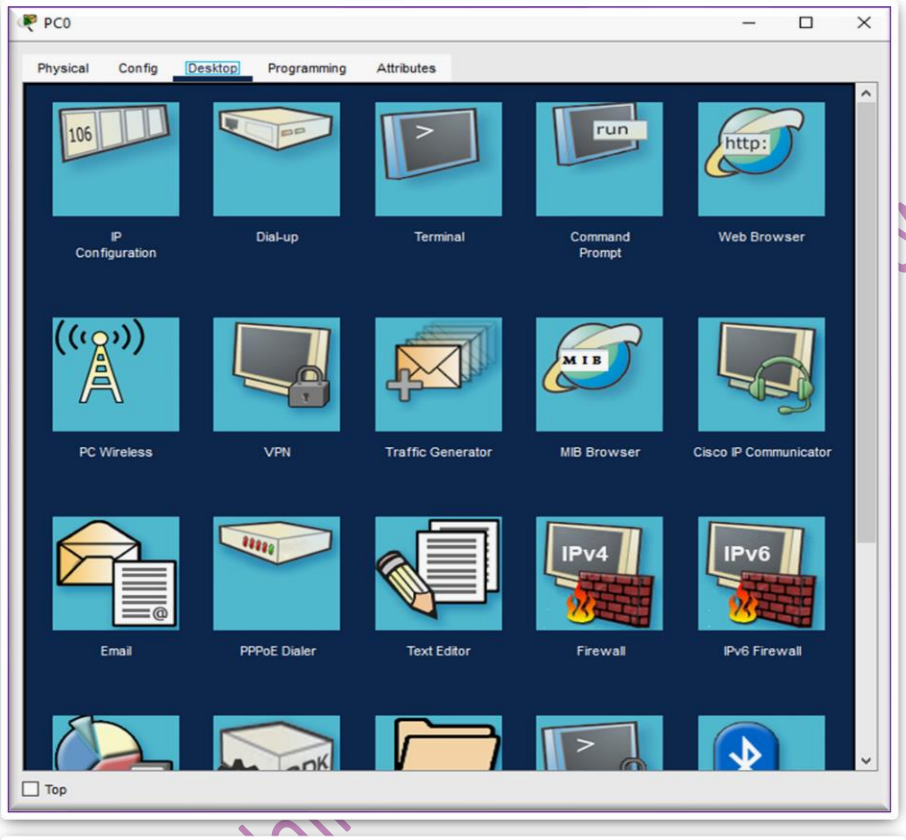
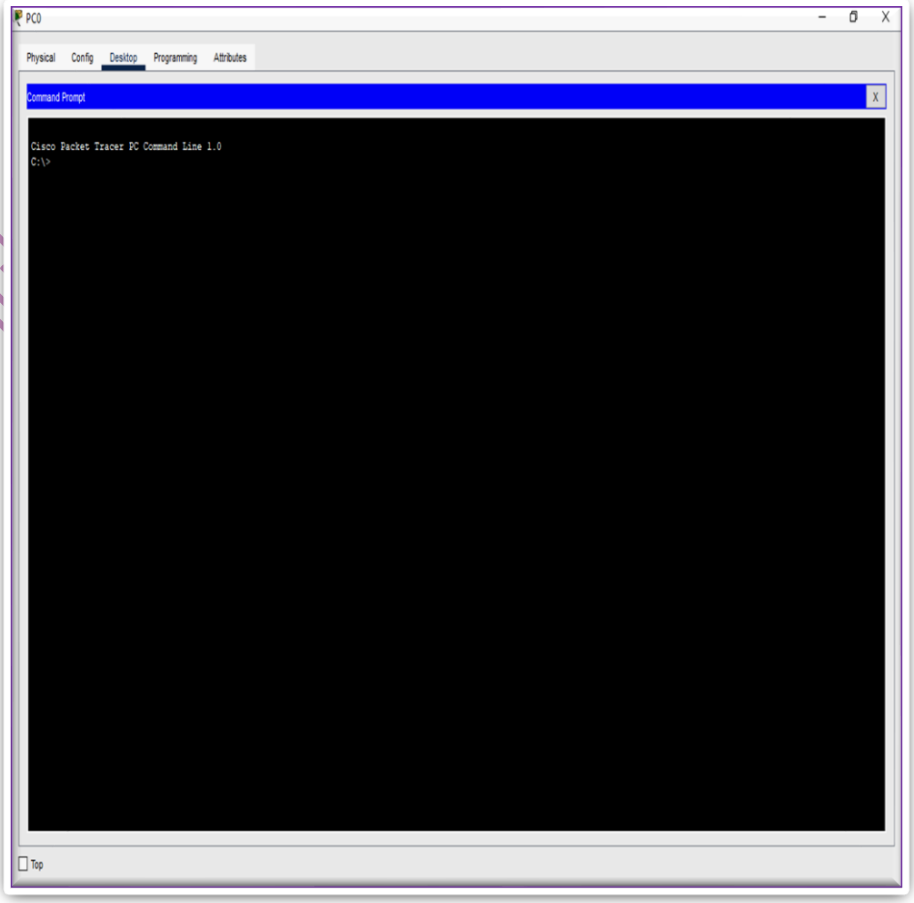
Network Testing Using CLI Commands

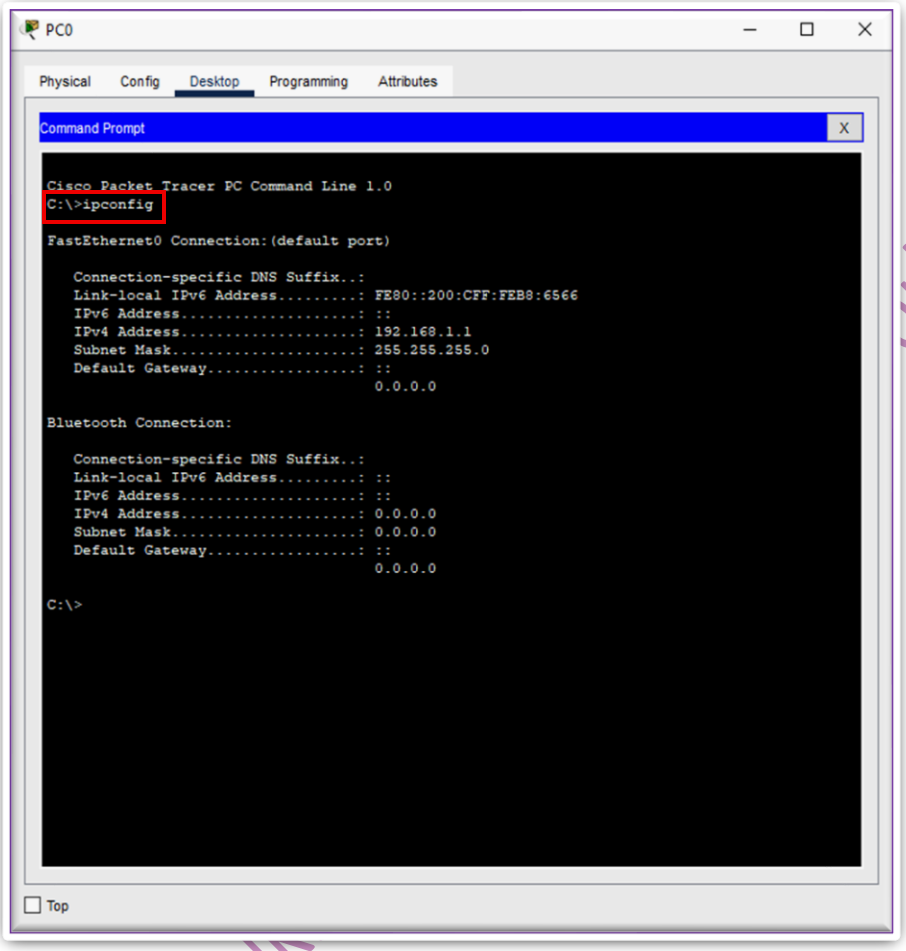
ipconfig

Open Command Prompt in first PC “PC0”

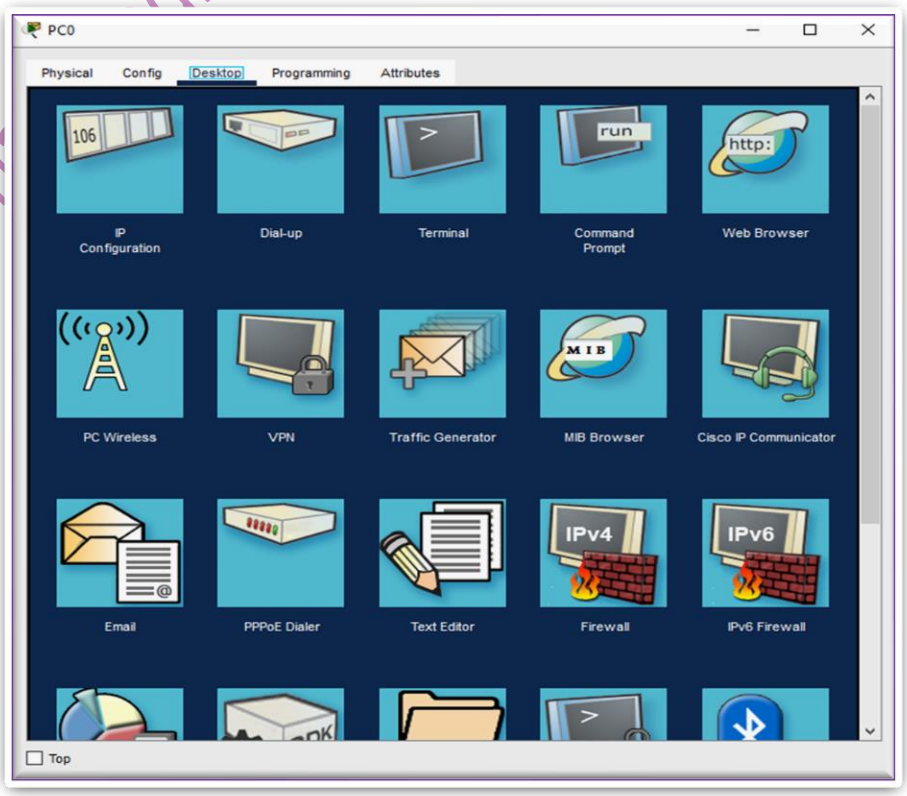




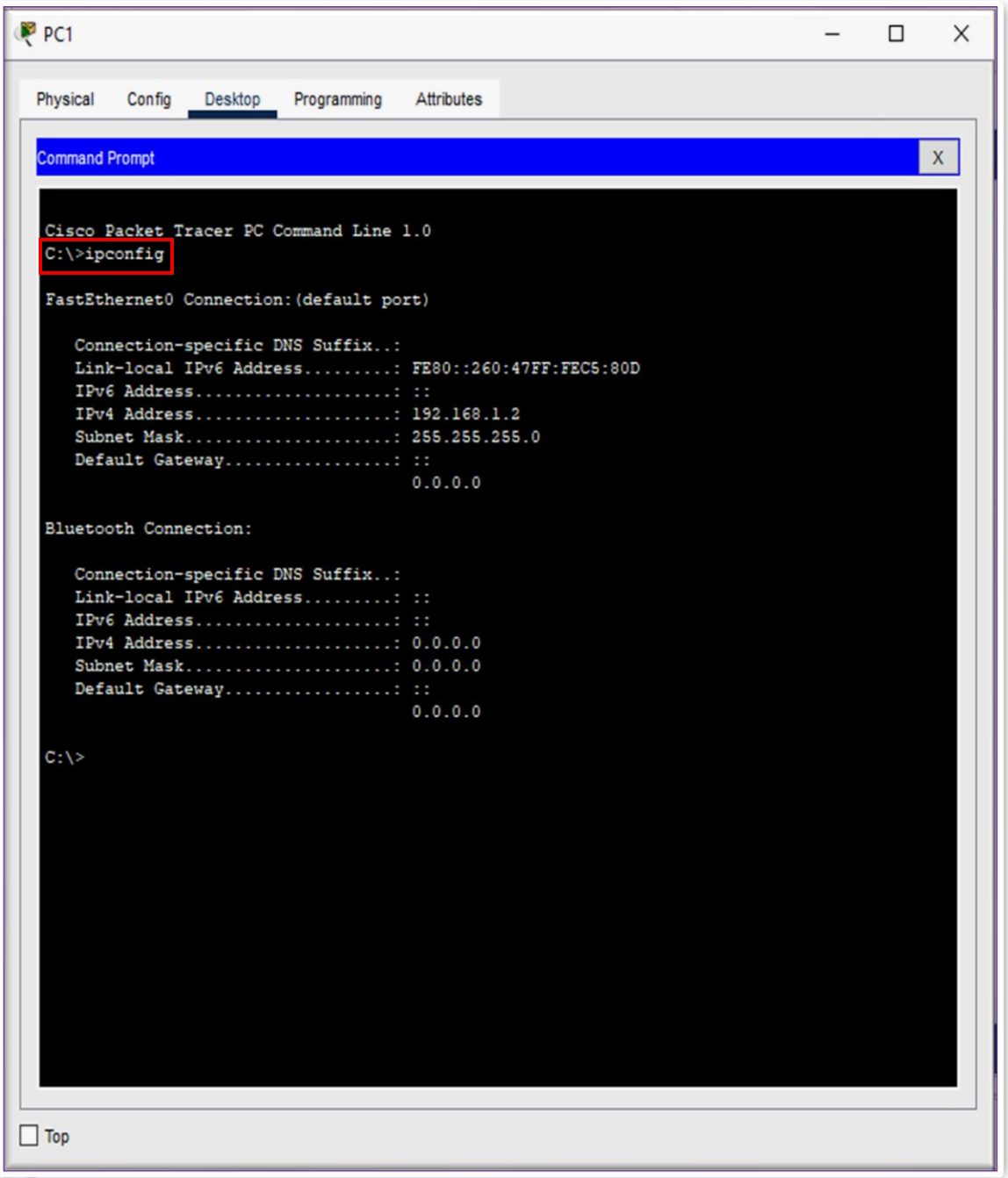
Display IP configuration of the PC



Open Command Prompt in second PC "PC1"



Display IP configuration of the PC

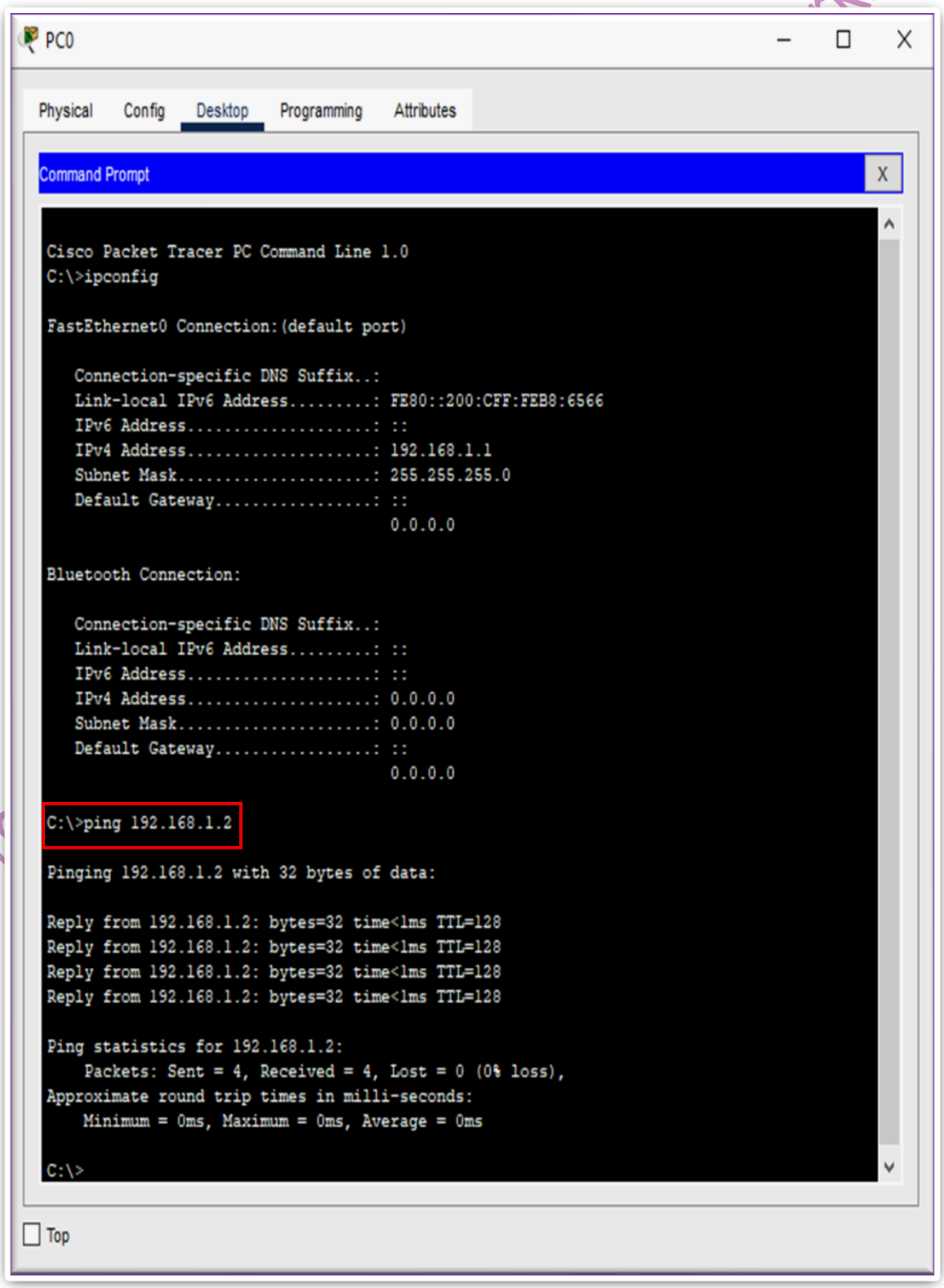


In this section we create computer network with command prompt. First open command prompt then run command "ipconfig" on both PCs do same process. After this process we move to the next procedure.

ping

Tests connectivity between two devices.

Ping in first PC “PC0”



Ping in Second PC “PC1”

PC1

Physical | Config | Desktop | Programming | Attributes

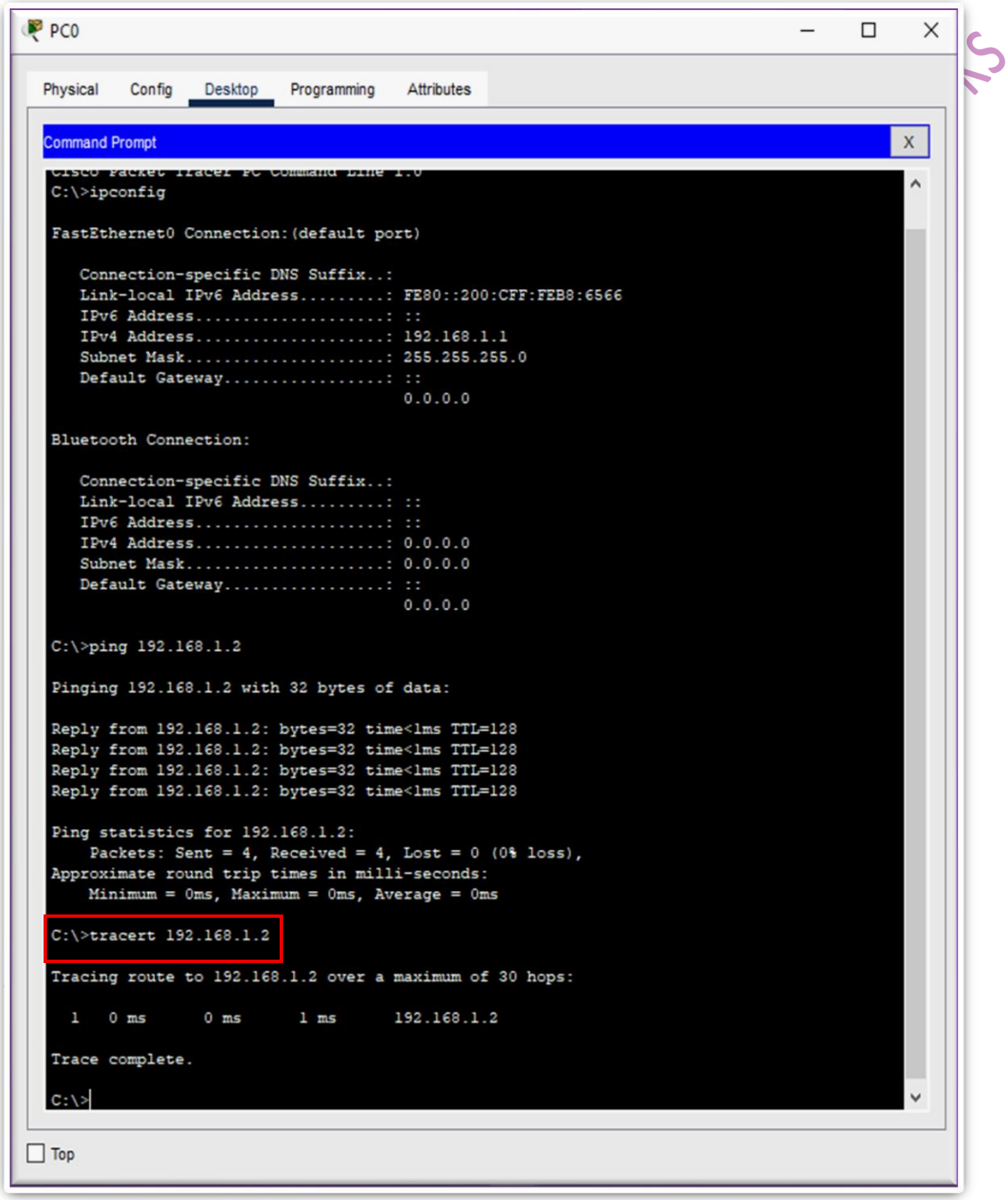
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:47FF:FEC5:80D
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Top

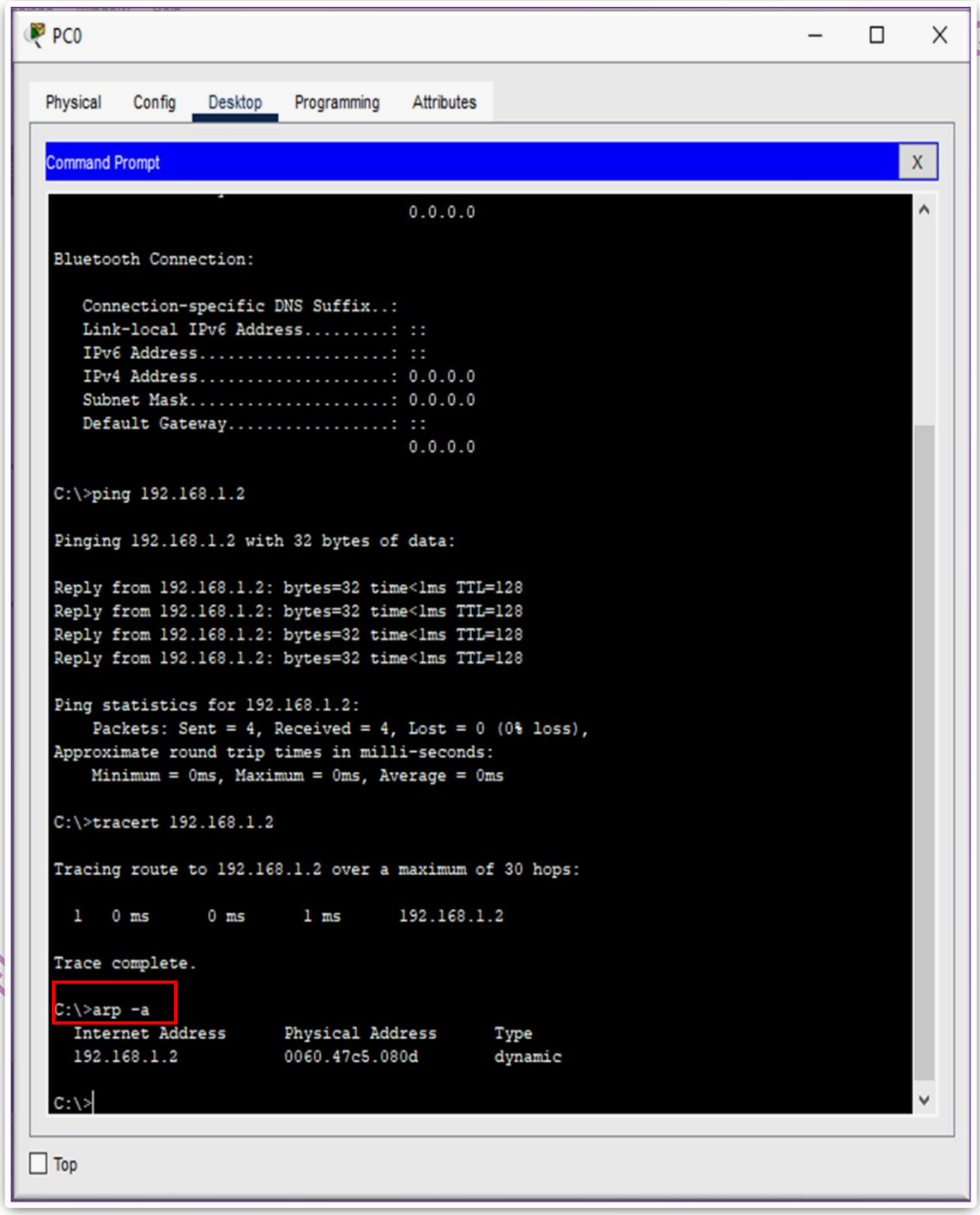
In this step we test connectivity between both two PCs. For this we run "ping" command on command prompt for both PC.

tracert

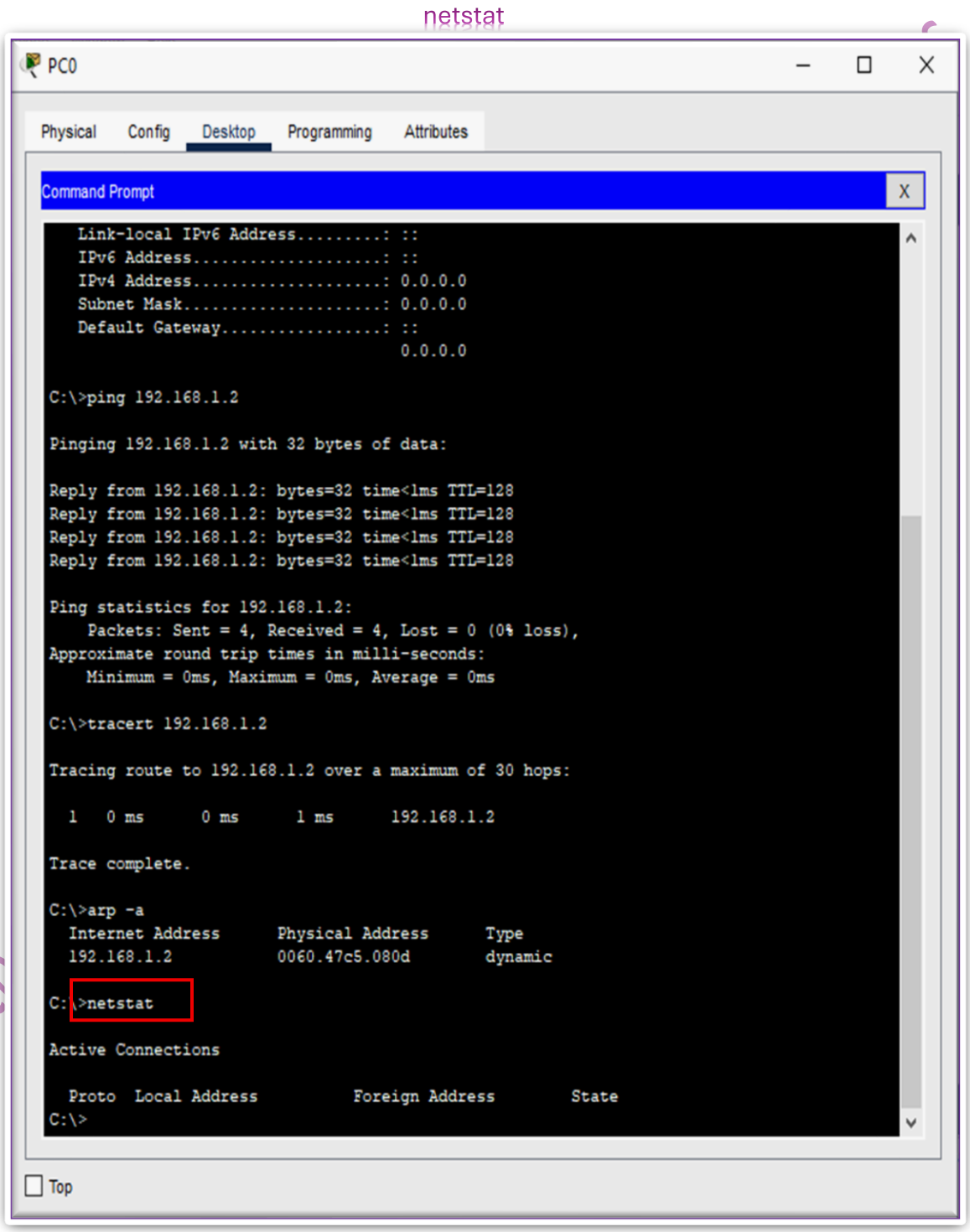


In this step we Shows the path taken by packets by using command prompt with command "tracert" from first PC "PC0". From this step we verify that both PCs are successfully connected through a switch.

arp -a



In this step we Displays ARP table (MAC address mapping) using command prompt with command "arp -a". We see the MAC address of the other PC after pinging.



Finally, we successfully created a network. And at last, we display active connections and statistics by command "netstat".

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